QP Code: _____ Reg. No: _____

Name : _____

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) MODEL QUESTION PAPER

FIFTH SEMESTER BCA HONOURS EXAMINATION

MG5DSEBCA301 – TIME SERIES ANALYSIS

(2024 ADMISSION ONWARDS)

Duration: 2 Hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse (An), Evaluate(E), Create(C), Skill(S), Interest(I)and Appreciation (Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

PART A
Short Answer Questions
Answer all questions. Each question carries 2 marks.

1.	Define a time series. Give any two examples.	[K]/ [CO1]
2.	State any four applications of time series analysis.	[K] / [CO1]
3.	What is a time series plot? State its importance.	[U] / [CO2]
4.	Define Weighted Least Squares.	[K] / [CO3]
5.	Define a Vector Autoregressive (VAR) model.	[K] / 'CO4]

[5 × 2 = 10]

PART B
Short Essay Questions
Answer any 5 out of 7 questions. Each question carries 6 marks.

6.	Explain the different types of data used in time series analysis. Distinguish between cross-sectional data and time series data with suitable examples.	[U] / [CO1]
7.	Define forecasting and state its major objectives and uses	[U] / [CO1]
8.	Explain the internal structures of a time series. Discuss trend, seasonal, cyclical and irregular variations.	[U] / [CO1]
9.	Explain the role of graphical displays and time series plots in the preliminary analysis of forecasting data.	[U] / [CO2]
10.	Explain the purpose of model adequacy checking. Describe the importance of residual analysis in regression models.	[U] / [CO3]
11.	Explain the basic architecture of a Neural Network used for time series forecasting. Describe the input, hidden and output layers.	[U] / [CO4]
12.	Define Spectral Analysis. Explain how a periodogram helps identify periodic or cyclical patterns in a time series.	[U] / [CO4]

[5 × 6 = 30]

PART C
Essay Type Questions
Answer any 2 out of 3 questions. Each question carries 15 marks.

13. Explain the general approach to time series modelling and forecasting in detail. Discuss the major stages involved from preliminary data analysis and model identification through parameter estimation, diagnostic checking, forecasting and evaluation of forecasting performance. [An] / CO2]

14. Discuss Regression Models for General Time Series Data in detail. Explain the problems that may arise due to autocorrelated errors and analyse the role of appropriate regression methods, including Generalized and Weighted Least Squares, in improving model estimation and inference. [An] / [CO3]

15. Explain the major approaches used in multivariate time series forecasting. Discuss Multivariate Stationary Processes, Vector ARIMA Models and Vector AR (VAR) Models. Analyse how Bayesian methods or Neural Networks can contribute to modern forecasting. [An] / [CO4]

[2 × 15 = 30]

——— End of Question Paper ———